

Seat No:

Enrollment No:

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech, Summer 2024-25 Examination

Semester: VI

Subject Code: 303105341

Subject Name: Cyber Security

Date: 25-04-2025

Time: 2.00 pm to 4.30 pm

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Define volumetric DDoS attacks. b) Mention one example of a volumetric DDoS attack. c) Explain one method to mitigate volumetric DDoS attacks.	06	CO4	BT1
	B. a) What is Wireshark, and how does it function as a network sniffer? b) Mention two primary use cases of Wireshark in vulnerability scanning. c) Explain how Wireshark helps detect suspicious network activity.	06	CO1	BT2
Q.2	A. What security dangers come with port forwarding?	04	CO1	BT4
	B. Explain what a Demilitarized Zone (DMZ) means in the context of firewalls.	05	CO4	BT1
	OR			
	B. How can regular software updates and patch management reduce security risks?	05	CO1, CO3, CO4, CO6	BT1
Q.3	A. How does what is a keylogger, and how does it work? apply in real-world situations?	04	CO3, CO4, CO6	BT2
	B. Define a "zeroday attack" within the context of computer crime?	05	CO5	BT4
	OR			
	B. What is the role of an Intrusion Prevention System (IPS) in a firewall?	05	CO3, CO4, CO6	BT2

SECTION - B

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Define the eradication phase in incident response. b) Define the recovery phase in incident response. c) Mention two key differences between eradication and recovery.	06	CO1, CO6	BT2

	B. a) Mention two common challenges faced during information system development. b) Explain one technical risk involved in developing an information system. c) Describe one organizational risk associated with information system development.	06	CO1	BT1
Q.2	A. Explain the role of Wireshark in network vulnerability scanning.	04	CO3	BT2
	B. How can vulnerability scanning tools such as Nmap be used to identify hidden services on a network?	05	CO3	BT3
	OR			
	B. Discuss the importance of regularly updating vulnerability scanning tools and techniques.	05	CO3, CO4, CO6	BT5
Q.3	A. Can you explain session hijacking and its effects?	04		BT6
	B. Write your view about increasing cybercrime and also suggest preventive measures.	05	CO6	BT3
	OR			
	B. Why is information security important for organizations?	05	CO1, CO3, CO4, CO6	BT3